

Prediction of Thermal Impedance Curves in Discrete Electronics Using Gated Recurrent Neural Networks

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Abstract—This paper proposes a Gated Recurrent Units (GRU)-based Neural Network trained on FEM-generated data to predict transient thermal impedance (Z_{th}) curves. This model achieves very high accuracy (~1% error) with millisecond-level prediction time; it combines the versatility of FEM with the efficiencies of RC models. Enhanced by positional encoding and custom loss function, this solution is scalable and fast enough for real time thermal analysis in chip design. While traditional design process may take days, the proposed method requires only a few hours for model preparation and training, delivering thermal impedance predictions in milliseconds.

Keywords—Semiconductor Devices, Thermal management, Thermal Impedance, FEM, GRU Neural Network.

1. Introduction

Heat management is becoming more and more important for today's discrete integrated circuits because of challenges of very high-power densities. Excessive heat always leads to performance degradation, reduction of life span, and eventually failure of the system. Designers must conduct accurate thermal characterizations, for predicting transient Z_{th} and surface hotspots evolution, in order to issue optimized layouts that ensure safer operational areas.

Finite Elements Method is one of the most conventional methods used to analyze thermal behavior in microelectronics. The method provides numerical solutions for complex heat transfer problems by discretizing the geometry into a mesh and solving locally the governing equations for each element. It can precisely treat complex material properties and boundary conditions, rendering it well-suited for chasing after realistic thermal behavior[1]. The scientific literature is abundant with works that analyze the finite element thermal management modeling of power integrated circuits (IC).

Often, to reduce computational time, engineers routinely employ equivalent circuit models such as the Foster and Cauer networks. Foster Networks reduce the characterization of the transient Z_{th} curve to a set of exponential terms, each being an RC branch, while Cauer networks series-connect RC stages, much like the physical thermal layers of the device. Though useful, these models need special tuning for each individual chip design, making them less efficient for design iterations [2].

Unfortunately, all yet classical available approaches to thermal modeling seem to have a trade-off between accuracy and efficiency, thus presenting a serious bottleneck in the design cycle. Trained through the results of a FEM simulation dataset, a neural model can quickly provide accurate thermal responses for novel chip designs

without programming a new simulation or re-calibrating. Neural networks can capture complex, non-linear relationships between input variables (e.g. geometry, materials, and time steps) and outputs (e.g., $Z_{th}(t)$), without explicitly encoding physical equations. Taking this path ensures a high-degree accuracy and fosters greater freedom in design exploration in significantly less time than traditional methods [3]. Thus, designers can run multiple configurations through parametric sweeps and analyze thermal behavior practically in real-time without further FEM runs or re-tuning RC networks. Previously, some works have also discussed the usage of machine learning for thermal modeling. For example, D. Shakar et al. [4] used Support Vector Regression to predict the steady-state behavior of silicon heat sources mimicking integrated circuits, attaining reasonable accuracy but limited scalability. Similarly, Liu [5] et al. used recurrent neural networks to model dynamic thermal behavior in high-power devices, proving that time-based models are capable of accurately describing heat flow over time. Nevertheless, the standard employed recurrent neural networks faced difficulty in the presence of long-term dependencies owing to vanishing gradients during training [6]. To overcome this problem, Salem et al demonstrated in [7] that GRUs are more suitable for time-series pattern learning [7], being based on gating mechanism to selectively update and memorize information in the hidden state.

The work in this paper aims to improve the Z_{th} modeling based on FEM through a data-driven solution coming from a neural network, namely GRU, to enable rapid but accurate thermal predictions (Z_{th} curves and the temporal behavior of surface temperature), compared to conventional approaches. The proposed methodology aims to efficiently (order of minutes) and accurately ($\epsilon_r \leq 1\%$ point to point) compute the Z_{th} for power semiconductor applications. The following sections will discuss the neural architecture, training procedure and performance assessment.

2. Background and Theoretical Framework

A. Thermal Impedance Computation

Numerical simulations, e.g. FEM, are a standard in the discrete electronics design for accurately modeling thermal behavior, for example to extract the transient thermal impedance curves (Z_{th}) and to estimate a time dependent temperature distribution[8]. The typical mathematical model solved is the transient heat conduction equation:

$$\rho c_P \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (1)$$

where:

ρ is the thermal density,
 c_p is the specific heat capacity,
 K is the thermal conductivity,
 T is the temperature field,
 Q is the internal heat generation per unit volume.

The transient thermal impedance indicates how quickly a discrete device develops a specific temperature to reach thermal equilibrium under a specified power load and helps engineers to quickly evaluate how any different design versions will affect their cooling performance[2]. The Z_{th} curve can be calculated using the equation:

$$Z_{th}(t) = \frac{T_h(t) - T_a}{P} \quad (2)$$

where:

T_h - Simulated heatsink temperature over time
 T_a - Ambient temperature (Simulation constrained)
 P - Applied power input used in simulation

An ideal discrete device model for computing thermal behaviour and for extracting the $Z_{th}(t)$ curves would comprise all integrated structure details, for example, bond wires, die-attach layers, solder joints, and PCB segments. However, FEM models are typically heavily simplified in order to obtain reasonable computing times, especially for designs that requires iterations over the designs or datasets. In addition, material properties are further homogenized to reduce the complexity of the problem [1].

As demonstrated in [9], neural networks can offer a faster alternative for extracting functional chip parameter such as R_{on} . In the same vein, thermal impedance computation can be accelerated by complementing FEM simulations with neural networks, particularly GRU. After training with data generated from different FEM results, the models can generalize new chip configurations and operating conditions, providing fast near-instantaneous predictions toward heat dissipation behavior. The working principles of this method are further presented.

3. Thermal Impedance Curve Prediction Using Gated Recurrent Neural Networks

The structure of a GRU is presented in Fig. 1 and its model is described in the following paragraph, where x_t denotes the input at the current time step and h_{t-1} represents the hidden state for the previous time step. The parameters W and U are learnable weight matrices for the input and hidden state, respectively, while b denotes the corresponding bias vectors. The function σ refers to sigmoid activation function and $\tanh$ is the hyperbolic tangent function, which is used to compute the candidate hidden state [7]. The symbol L represents a linear transformation (i.e., an affiliate operation such as $W \cdot x + b$), and C indicates vector concatenation used to combine input and hidden state information.

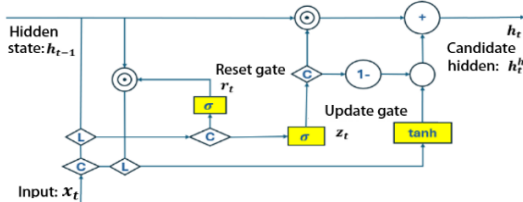


Fig. 1 GRU Cell Structure [7]



Fig. 2 Neural Network Architecture

GRU cells have two main gates: a reset gate that controls forgetting past information and an update gate that controls how much past information to keep. These gates calculate a temporary state, which then forms the new hidden state, enabling the model to learn long-term patterns effectively, important for application like thermal analysis.

The architecture of neural network layers is presented in Fig. 2, and explained in the text below:

1. *Sequence input layer* - Sequentially accept pre-processed and reshaped data.
2. *Normalization layer* - Employs layer normalization across features to stabilize training dynamics. This form of normalization mitigates internal covariate shifts within the network[10].
3. *Stack of GRU Layers* - Core architecture consists of 4 GRU layers, each given a configuration parameter in terms of number units ranging between 50 to 100 [7].
 - 3.1. *Exponential Linear Unit (ELU) Activation* - Each GRU block, except the one preceding the output layer, employs an ELU non-linearity in its pathway to develop stronger representations of the data.
4. *Fully Connected(Regression) Layer* - This architecture allows the network to provide a sequence of predictions for the original input timesteps through the sequence of GRU outputs mapped into the final prediction space.

4. Computational Methodology

The proposed approach is performed in two steps. In the first step, the training data is constructed with the assistance of FEM-based simulator. The schematic representation of chip structure and corresponding design parameters are presented in Fig. 3.

For simplification purposes, the chip structure is represented by four main components: the chip active area, the contact clip, the lead frame and the mold compound.

In the first iteration, the parameters of active area its width (W) and length (L) are modified function of the design rules. This variation spans from the minimum to the maximum permissible geometrical limits, using a fixed step size to ensure comprehensive coverage of all feasible configurations.

Although the other geometric parameters remain constant during this process, adjustments of the lead frame size and mold compound dimensions are necessary to maintain compliance with design rules and ensure functional model generation.

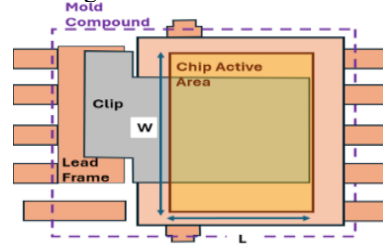


Fig. 3 Generic Chip Layout

These modifications are essential to preserve structural integrity and reflect realistic packaging constraints throughout the dataset generation.

For each design parameter set the thermal impedance curve $Z_{th}(t)$ is computed with the FEM simulator. Hence, a family of 600 $Z_{th}(t)$ curves have been computed as input for the GRU training step.

B. Data processing

Each geometrical parameter sets (W, L) and corresponding Z_{th} curve define the original input data which is subjected to a preprocessing phase.

The data preprocessing step ensures that the dataset is properly structured for sequential modeling by addressing both the temporal and statistical aspects necessary for deep learning. Through a series of transformations, the data is enriched with temporal context, cleansed of redundancy, and scaled to stabilize the training process.

Careful partitioning into training and testing subsets supports robust evaluation, while reshaping aligns the data with the expected input format of recurrent architecture of the GRU. Collectively these steps enhance the model's ability to capture and generalize from sequential data.

C. Custom Training Loop

The training loop for this application consists of a standard GRU training process enhanced with custom functions.

1) Standard training loop

The training process follows a standard approach for GRU-based sequence models. It starts with random initialization of the model's parameters and zeroing of moment estimates for adaptive optimization [7]. Each iteration includes a forward pass to generate predictions and compute loss, followed by a backward pass using backpropagation through time to capture temporal dependencies[7]. Parameter updates are handled by the Adam optimizer, which leverages running averages of gradients to ensure efficient convergence with minimal tuning[11].

To address limitations in learning from low-variation regions, the standard loop is extended with custom functions designed to enhance performance in saturated areas of the output sequence.

2) Custom Function

Custom functions are mathematically defined operations, integrated into the training process to improve performance by addressing specific behaviors such as saturation and gradient instability.

a) Detecting Saturated regions

The purpose of this operation is to check the region of the curves where the model output remains nearly constant for consecutive timesteps [9]. The saturated regions are areas where the output remains nearly constant over time, see Fig. 4.

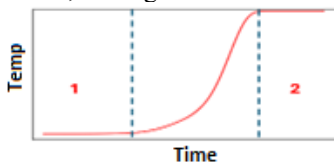


Fig. 4 The regions of Z_{th} curve: (1) first (2) second saturated regions

In these areas, the gradient of the predicted thermal impedance curve gets to zero.

$$dZ_{pred}/dt \approx 0 \quad (3)$$

where Z_{pred} represents the difference between the current and previous impedance predicted values, indicating minimal change over time.

This operation prevents the overfitting of the saturated areas and tries to compensate for the lack of response to dynamic input changes.

b) Gradient Smoothing

Since raw gradients fluctuate, a moving average filter smooths them across the update.

c) Custom Loss Function

The custom loss function penalizes saturated regions. The function considers a parameter for each saturated region(start-region/end-region). This function has an important role in enhancing prediction accuracy.

$$MSE(\hat{y}, y) = \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2 \quad (4)$$

$$L(\theta) = MSE(\hat{y}, y) + \alpha \sum_{t \in a} (\hat{y}_{t+1} - \hat{y}_t)^2 + \beta \sum_{t \in b} (\hat{y}_{t+1} - \hat{y}_t)^2 \quad (5)$$

where:

α -penalize parameter for the start-region

β -penalize parameter for the end-region

a-start region saturated domain

b-end region saturated domain

In practice, the final network loss ($L(\theta)$) is given by the normal Mean Squared Error (MSE) plus these two saturation penalties (scaled with the start-region, α and end-region, β penalize parameters).

After the training, the GRU network is able to predict in few milliseconds, the Z_{th} corresponding of any new transistor layouts, hence, providing a practical and scalable solution for thermal-aware circuit design.

5. Results and Discussion

The training data set for the first iteration of the predictor computation is obtained from 600 FEM simulations. All the other geometrical parameters remained constant besides the active chip area (2 varying parameters).

As shown in Fig. 5 prior to the integration of the custom loss function, the neural network exhibited important prediction errors.

The implementation custom loss function optimizes the fitting process improving, especially in the area of the last saturated region in Fig 4, where the gradient steady increases for each next predicted step. The simulated thermal impedance (Z_{th}) curve, and the corresponding predicted values produced through the neural network model (based on improved custom loss function algorithm), is presented in Fig.6.

The error between the predicted and simulated Z_{th} is about 1%.

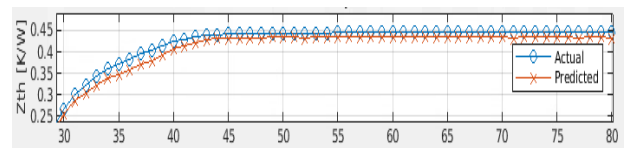


Fig. 5 Predicted Z_{th} Curve (using standard training algorithm) compared to the simulated one

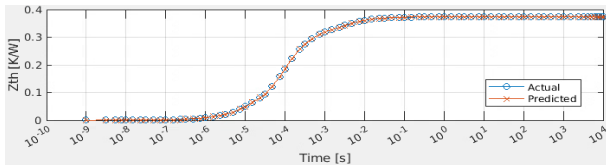


Fig. 6 Comparison between the predicted Z_{th} curve and simulated one

The proposed prediction method allows designers to efficiently analyze how thermal impedance changes function of the design parameters e.g. active area dimensions. The estimation method proposed in this paper is much faster and it does not require running a FEM simulation for each dimension of the active area.

Comparing the simulated and predicted Z_{th} curves in logarithmic scale reveals a slight deviation at the beginning of the curve (Fig. 7), but the impact is low since Z_{th} values are near zero in that region.

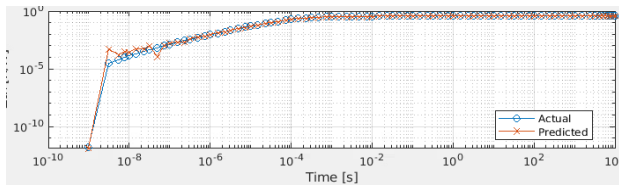


Fig. 7 Comparison between the predicted Z_{th} curve and the simulated one (Logarithmic Scale)

The Mean Square Error (equation 4), which serves as an indicator of how well the model fits the data, provides a useful metric to monitor and prevent potential overfitting during training, see Table 1.

Table 1 Evolution of MSE by the number of epochs

Epoch number	Mean Square Error
10	2.5E-02
50	3.4E-04
100	2.8E-04
300	1.8E-04
1000	4.1E-05
3000	9.3E-06

Based on the Table 1, the maximum MSE to validate the Z_{th} prediction model and avoid any potential overfitting was set to $1e-6$.

Although the training time is hardware dependent, for a single chip dataset model it takes between 10-20 minutes. In contrast, a single configuration FEM simulation takes approximately 20 minutes, whereas the trained predictor can generate a Z_{th} curve in a few milliseconds. Compared to the previously alternative methods, explained in the first chapter where the time to find the right chip configuration model could take several days, the prediction method developed in this paper can generate multiple Z_{th} curves rapidly, allowing designers to quickly validate the correct chip configuration model.

6. Conclusions and future work

A method for prediction transient thermal impedance (Z_{th}) curves based on a GRU neural network trained on FEM-generated data was presented. The usage of a thermal impedance predictor, enhanced through the integration of custom training functions, including saturated region detection, gradient smoothing, and a specialized loss function, increase model's ability to

capture thermal behavior especially for regions with minimal intensity change, where standard training typically underperforms.

By eliminating the need for time-consuming FEM simulations, the proposed method reduces computational time from minutes to milliseconds. This allows for a massive speed improvement while maintaining high accuracy with a relative error below 1%. Future work will focus on extending the model's applicability to a wider range of device geometries and integrating it into thermal-aware electronic design automation workflows.

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